

SPARWOOD, BC, Canada

WMO: 717820

Lat: 49.7450N

Lon: 114.8839W

Elev: 1138

StdP: 88.38

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-26.5	-23.1	-30.1	0.3	-25.9	-26.8	0.4	-22.9	9.4	-11.4	8.0	-10.6	2.3	10	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	29.8	15.8	27.9	15.2	25.9	14.5	16.9	26.5	16.0	25.4	15.1	24.2	3.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	11.2	18.8	12.5	10.4	17.9	11.5	9.7	17.3	51.7	26.6	48.7	25.5	46.1	24.3	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.4	5.7	-31.1	32.6	3.0	2.0	-33.2	34.0	-35.0	35.1	-36.7	36.2	-38.9	37.7		
(5)			-31.3	18.8	3.1	1.6	-33.5	19.9	-35.3	20.9	-37.0	21.8	-39.2	22.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-6.5	-5.1	-0.3	4.3	8.9	12.5	16.6	15.7	10.8	4.3	-1.9	-7.1		
	DBStd	9.57	7.56	6.32	5.24	3.58	3.61	3.09	3.01	2.94	3.62	4.11	6.00	6.97		
	HDD10.0	2604	511	423	319	174	64	11	1	1	33	179	358	529		
	HDD18.3	5109	769	657	577	422	292	177	70	91	226	434	608	788		
	CDD10.0	561	0	0	0	2	32	86	205	176	57	4	0	0		
	CDD18.3	25	0	0	0	0	0	1	16	8	1	0	0	0		
	CDH23.3	1041	0	0	0	1	23	84	485	382	67	0	0	0		
	CDH26.7	289	0	0	0	0	2	17	152	109	10	0	0	0		
	WSAvg	2.4	2.5	2.5	2.7	2.5	2.3	2.2	2.2	2.1	2.1	2.3	2.5	2.3		
	PrecAvg	859	83	59	67	53	77	86	47	45	59	88	102	90		
(15) Precipitation	PrecMax	1180	144	114	165	93	149	184	143	147	176	157	264	173		
	PrecMin	657	12	15	25	29	27	54	0	13	8	42	16	47		
	PrecStd	149	30	29	36	17	32	32	35	30	39	38	72	32		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	8.5	14.2	21.2	25.9	29.1	32.6	31.9	28.2	19.9	12.1	6.0		
		MCWB	4.6	3.5	6.0	10.1	13.1	15.7	17.1	15.5	14.7	11.4	6.6	4.1		
	2%	DB	5.0	5.9	10.5	17.7	23.3	25.7	30.4	29.5	25.3	16.4	9.3	4.2		
		MCWB	2.6	2.4	4.7	8.7	11.9	14.5	16.0	15.5	13.7	9.5	6.4	2.3		
	5%	DB	3.6	4.3	8.4	14.6	20.6	23.4	28.3	27.6	22.6	13.7	7.3	3.0		
		MCWB	1.6	1.6	3.9	6.8	11.0	13.2	15.6	15.0	12.8	8.1	4.7	1.4		
	10%	DB	2.4	2.8	6.5	11.7	17.7	20.9	26.2	25.4	19.6	11.3	5.5	1.7		
		MCWB	0.8	0.6	2.9	5.5	9.7	12.4	14.9	14.4	11.5	6.8	3.4	0.3		
	0.4%	WB	5.0	4.5	7.1	10.8	14.0	16.7	18.6	17.8	15.7	12.3	8.5	4.5		
		MCDB	7.1	6.7	11.5	19.4	23.5	26.5	29.1	27.7	23.9	17.9	10.5	5.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	3.0	3.1	5.5	9.0	12.7	15.3	17.3	16.5	14.2	10.2	6.5	2.7		
		MCDB	4.6	5.3	9.3	16.8	21.3	24.1	26.9	26.4	23.4	15.1	8.9	3.7		
	5%	WB	1.9	1.8	4.4	7.4	11.5	14.0	16.4	15.6	13.1	8.7	5.0	1.5		
		MCDB	3.2	3.9	7.9	13.4	19.0	21.7	25.7	25.4	21.5	13.0	7.0	2.8		
	10%	WB	0.9	0.7	3.2	6.0	10.2	12.8	15.5	14.8	11.9	7.2	3.6	0.4		
		MCDB	2.2	2.6	6.1	11.1	16.8	19.2	24.4	24.0	18.9	10.7	5.5	1.7		
	5% DB	MDBR	8.1	10.0	9.7	11.7	13.1	13.2	16.4	16.9	14.7	10.9	7.6	7.3		
		MCDBR	7.0	9.7	12.6	18.7	20.0	19.4	21.0	22.0	21.6	15.8	8.8	5.7		
		MCWBR	5.3	6.8	7.7	10.3	9.9	9.0	8.6	9.3	10.8	9.3	6.1	4.6		
		MCWBR	6.4	8.7	11.5	17.0	17.9	17.2	18.6	19.4	20.1	14.6	8.1	5.7		
(35) Mean Daily Temperature Range	5% WB	MCWBR	5.1	6.5	7.3	9.6	9.1	8.2	8.1	8.8	10.3	9.0	6.1	4.9		
	taub		0.246	0.258	0.283	0.328	0.345	0.333	0.336	0.344	0.312	0.292	0.267	0.238		
		taud	2.361	2.334	2.315	2.271	2.389	2.476	2.482	2.461	2.532	2.565	2.488	2.389		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		825	901	932	916	913	924	916	892	893	858	799	787		
		Edh at Noon	63	84	103	121	113	105	103	100	84	68	57	53		
	RadAvg		1.10	2.07	3.12	4.49	5.31	5.71	6.67	5.49	3.82	2.29	1.16	0.86		
(45) All-Sky Solar Radiation	RadStd		0.10	0.16	0.28	0.35	0.45	0.47	0.53	0.40	0.41	0.24	0.11	0.07		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page